

- In this course we will consider problems in four different ways, referred to as the **Rule of 4**:
 - Analytical / Algebraic
 - Graphical / Geometrical
 - Numerical
 - Verbal
- Functions are the foundation of Calculus
*A **function** is a rule that assigns to each value of x exactly one value $f(x)$.*
*Above x is the **independent** variable, and if $y = f(x)$, then y is the **dependent variable**.*
- MA 130 – Calculus I is the study of functions of a single independent variable.
- Some of the functions we will encounter this semester can't be characterized, but many will be types of functions you might have seen before:
 - Polynomial
 - Trigonometric
 - Exponential
 - Logarithmic
 - Piecewise
- We can represent functions in many ways:
 - Algebraic definition
 - Graph on the Cartesian Coordinates
 - Table
 - Set of ordered-pairs
- The set of values of the independent variable is called the **domain**, and the set of all possible values of the function is the **co-domain**, while the set of values actually attained by the function is called the **range**.
- A function assigns a unique value of $f(x)$ to x , which leads to the **vertical-line** test.
- What is an example of the fourth **Rule of 4**, verbal?
 "I went to class, then I went right back to my room."

Graphically, this might look like:

